

Qianwei(Robin) Wang

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EDUCATION

B.S. Computer Science, [University of Michigan-Ann Arbor](#), MI Sep.2023-May.2026
GPA: 3.92/4.0
B.Eng Electrical Information Engineering, [Xidian University](#), China Sep.2021-Jul.2023
GPA: 3.90/4.0

SKILLS

Programming: C/C++, Python, Matlab
Tools: ROS, PyTorch, OpenCV, Eigen, Ceres

EXPERIENCE

Summer Intern, [Advanced Agent-Robotics Technology Lab](#), Carnegie Mellon University
Advisor: Dr. Katia Sycara May.2025-now
Undergraduate Research Assistant, [LIVE](#), University of Michigan
Advisor: Dr. Vineet R. Kamat Sep.2023-now
Summer Intern, [ARM](#), University of Michigan
Advisor: Dr. Dmitry Berenson May.2024-Aug.2024

PUBLICATION

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- ① **Wang, Qianwei**; Xu, Yifan; Kamat, Vineet; Menassa, Carol. "[OVAMOS: A Framework for Open-Vocabulary Multi-Object Search in Unknown Environments](#)" (in submission)
② Xu, Yifan; Luo, Ziming*; **Wang, Qianwei***; Kamat, Vineet; Menassa, Carol. "[Point2Graph: An End-to-end Point Cloud-based 3D Open-Vocabulary Scene Graph for Robot Navigation](#)." (accepted by **ICRA 2025**)
③ Xu, Yifan; **Wang, Qianwei**; Kamat, Vineet; Menassa, Carol. "[Socially-Aware Shared Control Navigation for Assistive Mobile Robots in the Built Environment](#)." (Accepted by **Journal of Computing in Civil Engineering**)

SELECTED PROJECTS

Neuro-Symbolic Bilevel Learning Framework for Bilevel Planning
Supervised by Dr. Katia Sycara, CMU May.2025-now
Working on improving learning neural predicates directly from demonstrations.
Related paper: [Bilevel Learning for Bilevel Planning](#)
Open-Vocabulary Multi-Object Search in Unknown Environments①
Supervised by Dr. Vineet R. Kamat, [LIVE](#), University of Michigan Nov.2024-now
I am working on enabling the robot to simultaneously build beliefs about **multiple types of objects** while exploring the environment.

Socially-Aware and Shared Autonomy-based Wheelchair^②

Supervised by Dr. Vineet R. Kamat, [LIVE](#), University of Michigan

Sep.2023-now

We have designed a navigation framework for wheelchairs that integrates **user preferences** into both the local planner and the global planner. In addition to basic obstacle avoidance capabilities, we developed a **SS (Socially Aware)-MPC-DCBF (Dynamic Control Barrier Function)** framework to enable the wheelchair to effectively and safely **avoid dynamic pedestrians**.

My contribution:

© I utilized Gazebo to **establish a simulation environment** and **built an experimental platform** on a wheelchair equipped with a LiDAR, camera, IMU, and CAN communication system in real-world settings. Additionally, I **implemented state-of-the-art odometry and mapping algorithms**, such as FASTER-LIO, LEGO-LOAM, and RTAB-Mapping, to evaluate and enhance the system's performance in both simulated and real-world environments.

© In the physical-world experiments of our research, I was responsible for transferring the algorithms from the simulation environment to the real robot. This process included using **PID** control to compensate for errors in the wheelchair's actual motion and fine-tuning the parameters of the **MPC-based local planner**.

Tool Retrieval in Agricultural Environments

Supervised by Dr. Dmitry Berenson, [ARM](#), University of Michigan

May.2024-Aug.2024

This project aimed to enable the Spot robot to operate in **outdoor** agricultural environments by responding to human instructions. The robot was designed to navigate to the vicinity of the specified tool, identify the target tool from a **cluttered set of tools** in an **open-vocabulary** setting, and perform the pick-up and subsequent delivery to the user's hand.

My contribution:

During the experiments, I identified that the current open-vocabulary detector performed poorly in scenarios involving **long distances or objects in cluttered environments**. To address these challenges:

© For long-distance detection, I adopted a strategy inspired by VLFM (Vision-Language Frontier Map), leveraging the scene reasoning capabilities of Vision-Language Models (VLMs) to guide the robot to regions where the target object was likely to be located.

© For object-in-clutter scenarios, I designed a solution combining a **real-time detector (YOLO-World)** with a **dynamic scanning** approach, enabling the robot to collect **multiple viewpoints** and improve detection accuracy.

3D Open-Vocabulary Scene Graph for Robot Navigation^③

Supervised by Dr. Vineet R. Kamat, [LIVE](#), University of Michigan

Mar.2024-Sep.2024

We developed Point2Graph, a point cloud-based 3D open-vocabulary scene graph framework for robot navigation that **eliminates reliance on RGB-D images** while integrating room and object detection with open-vocabulary classification.

My contribution:

© My contribution lies in identifying the limitations of learning-based room segmentation methods in complex real-world scenarios. To address this, I incorporated room boundary information derived from further processing point cloud data, which was then used for **denoising the scene density map**, achieving improved segmentation performance.

AWARDS

First Prize in The Chinese Mathematics Competitions

Second Prize in University Championship-RoboMaster (held by DJI)-[RMUC](#)

Second-class Scholarship at the Xidian University